Final Exam PHYS530 (June 06, 2022)

Advanced electrodynamics 1

1. **(50 points) Charge in constant E&B fields.** Consider the motion of charged particle under the magnetic field in z-direction and electric field in the yz-plane. The Lorentz force can be expressed as $F = e\vec{E} + \frac{e}{c}\vec{v} \times \vec{H}$.

   a. (10) Set up the equations of motion for x y, z directions.

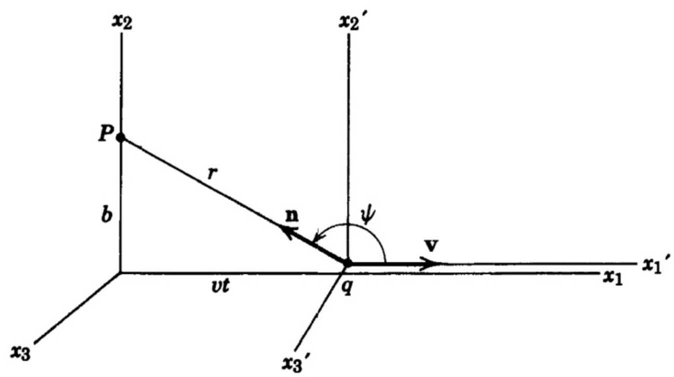
   b. (10) Solve the different equation obtained above to find the velocities.

   c. (10) Obtain the expression for the frequency and find the drift velocity.

   d. (20) Find the time-evolving paths of the particle, and draw the particle trajectories for three different initial conditions of $(1)\dot{x}(t=0)=0, (2)\dot{x}(t=0)>0$ & $(3)\dot{x}(t=0)<0$. For all three initial conditions, $\dot{y}(t=0)=0$ & $y(t)\geq 0$.

2. **(50 points) Fields by a moving charge**.

   The fields at P for the moving charge as shown in the diagram below.



   a. (20) First, find the field by an observer at P moving with the charge (fields in the moving frame). The origin coincides at time zero.

   b. (20) Using the Lorentz transformation matrix, obtain the fields at P in the rest frame. Use the Lorentz transformation relations below.

$$\begin{aligned} E_1' &= E_1 & B_1' &= B_1 \\ E_2' &= \gamma(E_2 - \beta B_3) & B_2' &= \gamma(B_2 + \beta E_3) \\ E_3' &= \gamma(E_3 + \beta B_2) & B_3' &= \gamma(B_3 - \beta E_2) \end{aligned}$$

   c. (10) Plot the field strength (for E_2) for fast moving charge ($\beta \sim 1$), and sketch the field lines.